

Name: _____

Date: _____

INEQUALITIES, (INEQUALITIES IN TWO VARIABLES)

LO: I can represent and identify regions defined by linear inequalities in two variables.

Inequalities with x and y

An **inequality in two variables** such as $y < 2x + 1$ defines a **region** on a coordinate grid, not just a line.

Strategy: Draw the boundary line (dashed for $<$ or $>$, solid for $\leq$ or $\geq$), then shade the correct region by testing a point.

Quick examples

●	$y < x + 3$: region below the line $y = x + 3$	dashed line
●	$y \geq 2x - 1$: region above and including the line	solid line
●	$x + y \leq 5$: test (0,0) gives 0 $\leq$ 5, true, so shade that side	test point method
●	$y < x + 3$ uses a solid boundary line	strict inequality = dashed

Key idea

Draw the boundary line, choose dashed or solid, then test a point to determine which side to shade.

$$y \bigcirc mx + c$$

solid line, shade below

Solid line includes the boundary; shade the correct side

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - identifying the region

Describe the region $y < x + 2$

Boundary line: $y = x + 2$ (dashed, as strict $<$)

Test (0, 0): $0 < 0 + 2 = 2$, true

Region: below the dashed line $y = x + 2$

Below dashed line $y = x + 2$

Use a test point to verify which side satisfies the inequality.

Example 2 - solid boundary

Describe the region $y \geq 3x - 1$

Boundary: $y = 3x - 1$ (solid, as $\geq$)

Test (0, 0): $0 \geq -1$, true

Region: above and on the solid line

Above/on solid line $y = 3x - 1$

Solid line means points ON the line are included.

Example 3 - testing a point

Is the point (2, 5) in the region $x + y \geq 6$?

Substitute: $2 + 5 = 7$

Check: $7 \geq 6$?

No, $7 > 6$, so (2, 5) is NOT in the region

No

Substitute the coordinates into the inequality and check if it is true.

Example 4 - combined region

Describe the region satisfying $y > 1$, $x < 4$, and $y < x$

$y > 1$: above the dashed line $y = 1$

$x < 4$: left of the dashed line $x = 4$

$y < x$: below the dashed line $y = x$; the overlap is the triangular region

Triangular region above $y = 1$, left of $x = 4$, below $y = x$

The feasible region is where all three inequalities are satisfied simultaneously.

YOUR TURN – PRACTICE (deliberate practice)

1. Identifying boundary lines

State the boundary line and whether it is dashed or solid.

- a) $y < 3x + 1$
- b) $y \geq x - 4$
- c) $x + y \leq 10$
- d) $2x + y > 6$

2. Test point method

Test (0, 0). Is it in the region?

- a) $y < x + 5$
- b) $y > 2x + 3$
- c) $x + y \leq 4$
- d) $3x - y \leq 1$

3. Describing regions

Describe the region (above/below, left/right of boundary).

- a) $y > 2$
- b) $x \leq 5$
- c) $y < -x + 3$
- d) $y \geq \frac{1}{2}x$

4. Checking points

Determine if the given point satisfies the inequality.

- a) (1, 4) in $y \geq 3x + 2$?
- b) (3, 1) in $y > x - 1$?
- c) (2, 3) in $x + y < 6$?
- d) (-1, 2) in $y \geq 2x + 5$?

5. Combined regions

Describe the region satisfying all given inequalities.

a) $y \geq 0, x \geq 0, x + y \leq 4$

b) $x > 1, y < 5, y > x$

c) $y \geq x + 2, y \leq 0, x \leq 3$

d) $x + y \leq 8, x \geq 2, y \geq 1$

e) $y < 2x, y > -1, x < 3$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) A strict inequality ($<$ or $>$) uses a dashed boundary line ☐

Correct answer: _____

Reason: _____

b) $y < x + 3$ means the region above the line $y = x + 3$ ☐

Correct answer: _____

Reason: _____

c) Testing (0, 0) tells you which side of the line to shade ☐

Correct answer: _____

Reason: _____

d) $x \geq 2$ and $y \geq 1$ both use solid lines because the boundary is included ☐

Correct answer: _____

Reason: _____

e) $y \geq 2x$ means the region below the line $y = 2x$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A factory makes chairs (x) and tables (y). Each chair needs **2 hours** and each table needs **3 hours**. They have **30 hours** available. Write an inequality for this constraint and identify three integer points (x, y) that satisfy it, where $x \geq 0$ and $y \geq 0$.

Expression: _____

Simplified: _____

b) Find the area of the region satisfying $x \geq 0, y \geq 0$, and $x + y \leq 6$. What shape is this region? If we add $y \geq x$, what is the new area?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

